Data Science – Mini Project 3 – Sami Seppälä

Introduction

Using data from collected from wikiart.org the aim is to create a network using networkx in python and answer the following questions:

1. Which were the most influential artists?

2. Which were the most influential movements?

3. Which were the most influential institutions?

4. Which nationalities concentrate the majority of artists?

5. Which are the biggest communities in the network?

The data consists of 4 csv files containing the following information:

artists.csv: URL of the artist at WikiArt, id, image URL, nation, name, total of art work, interval of active years.

relationships.csv: URL of the artist at WikiArt, list of friends, list of artist that they were in influenced by, list of artist that they influenced, list of art institutions that the artist

studied, list of schools that was part of, type (artist or collection).

institution.csv: City, country, name, URL of the institution at WikiArt.

schools.csv: name, school URL at WikiArt.

Methodology/Results

Since the goal is to mainly analyze relationships between different variables in the tables I chose not to do anything with the missing values. As centralities will be calculated this is also not needed.

Employing EDA I made the following visualizations from the data:

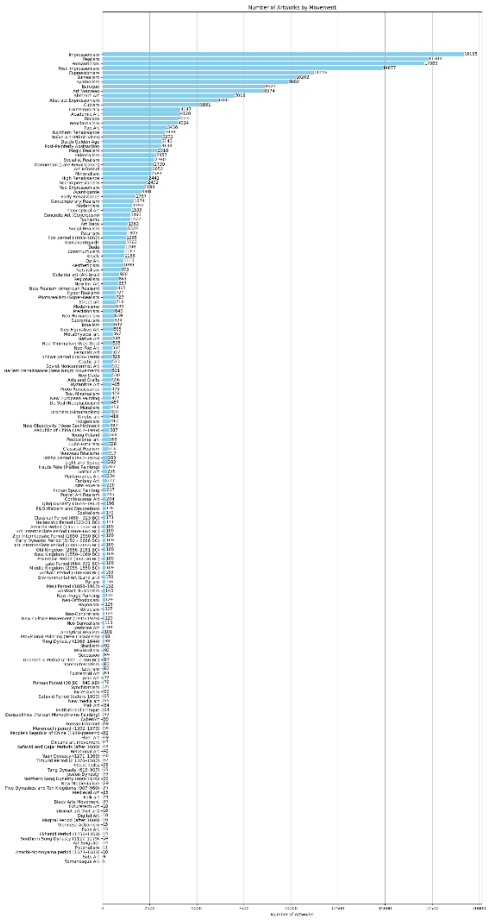


Figure 1 - Artworks by movement

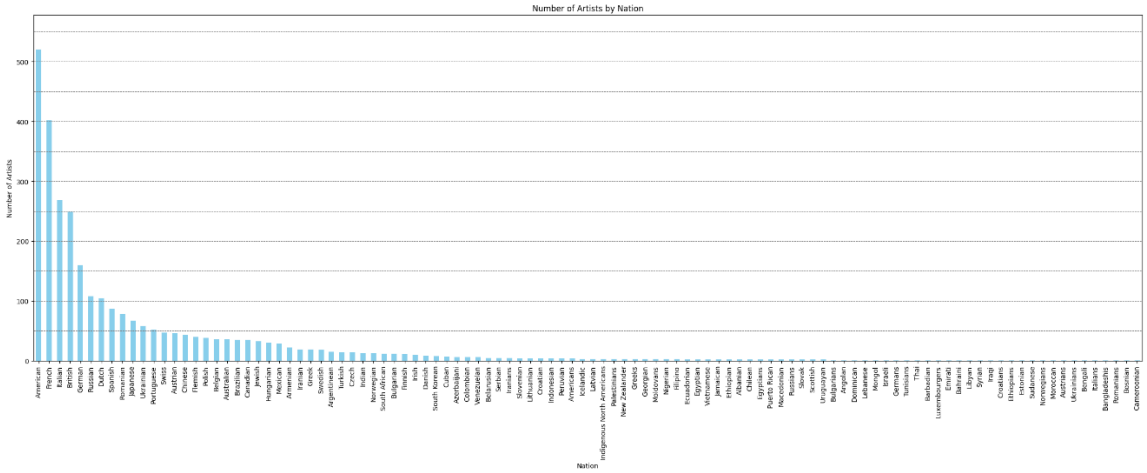


Figure 2 - Artists by school

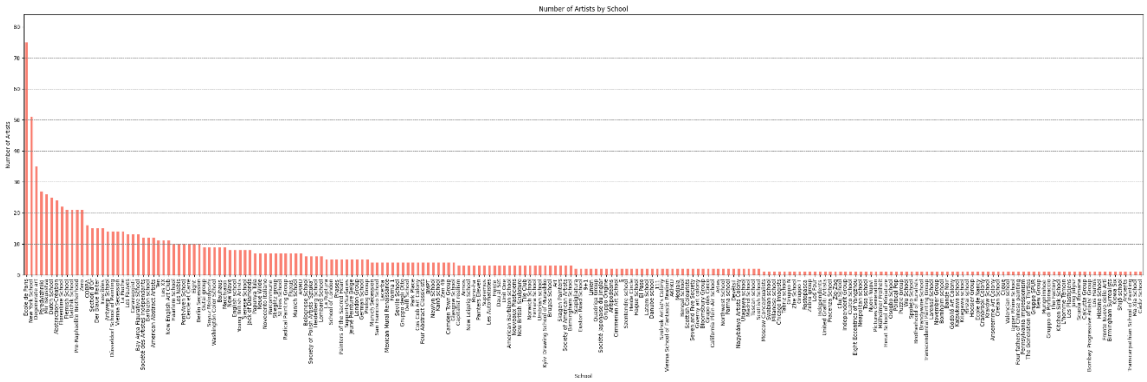


Figure 3 - Artists by school

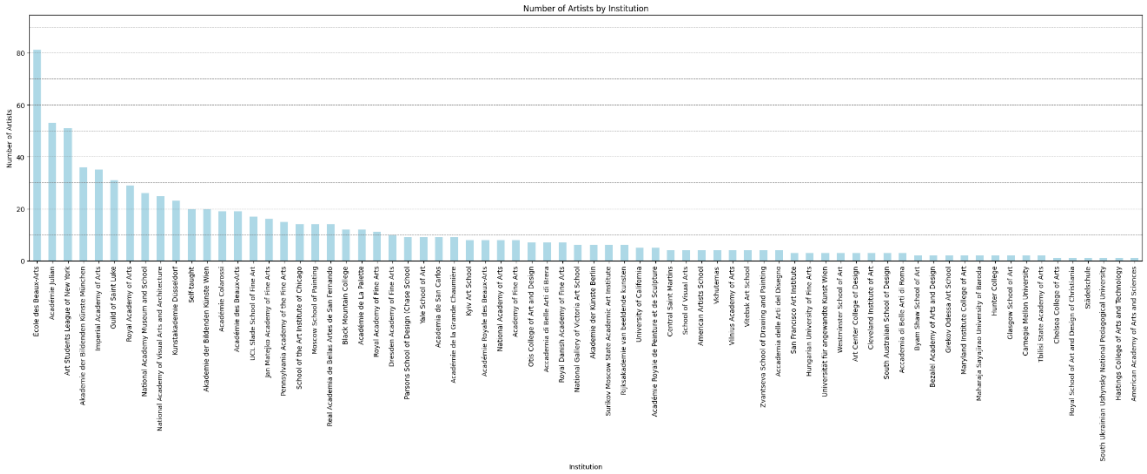


Figure 4 - Artists by institution

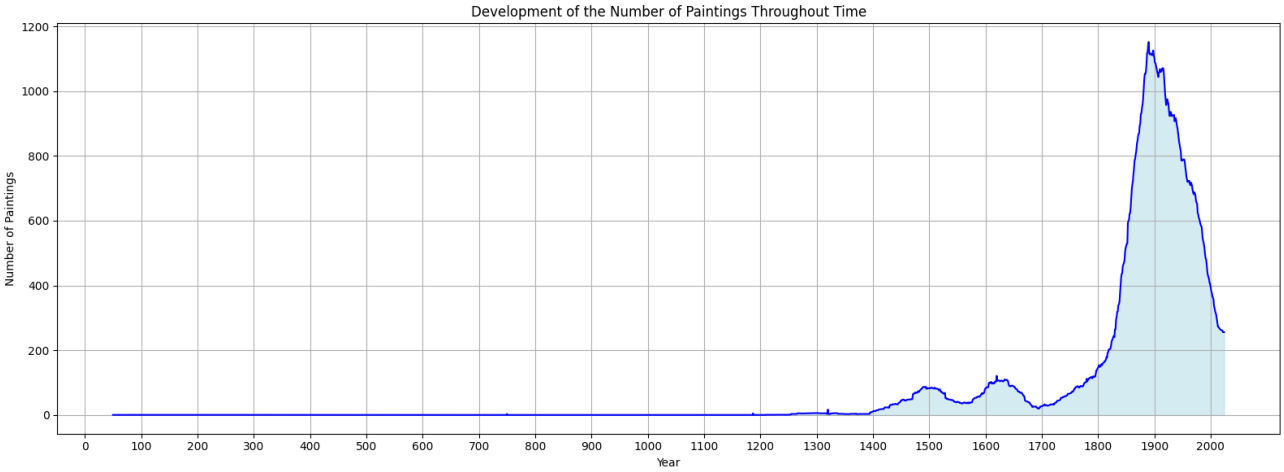


Figure 5 - Number of artworks trough time

Figure 1 shows that impressionism, realism and romanticism have the highest number of artworks. This is supported by Figure 5 that show the development of the number of artworks trough time where the most can be seen at the end of the 19th century. Figure 5 is calculated using the given time periods when artists were active and dividing it by the number of artworks. This means that long time periods with very few artworks get divided close to 0 i.e. Ancient Egypt etc. Figure 2-4 show artists by nation, school, and institution. The top nation is America, top school is École de Paris and top institution is École e des Beaux-Arts, all by number of artists.

I started by creating a network that included all the nodes and edges I would be needed trough the different networks and would be later used for the community analysis. The nodes were artists, institutions, movements, and nationalities, while the edges consisted of the nationalities of artists, influenced_by, influenced_on, where they studied, and which movement they were a part of. After creating the network, centralities were calculated for the pyvis interactive visualization where sizes of nodes are scaled according to their centrality. Some of the most prominent nodes consisted of nations and movements.

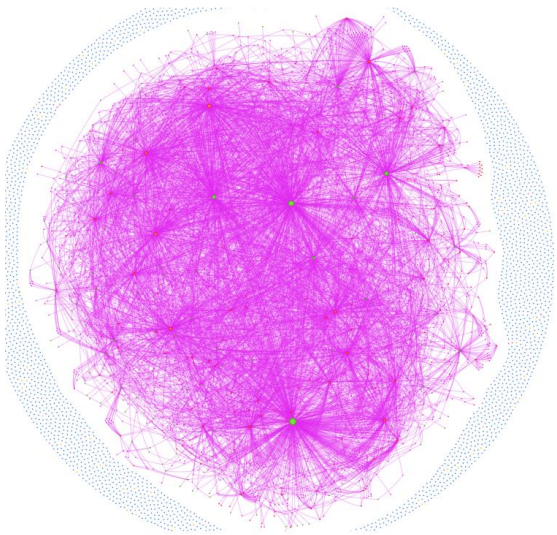


Figure 6 - Pyvis visualization

In order to answer questions 1-5, I created new networks with specific nodes and edges specifically needed to answer them. For the artist network the nodes consisted of artists and the edges of influenced_on. The centrality was

then calculated to find the most infuential artists, where Pablo Picasso had the most influence:

```
Most Influential Artists:
                Artist  Artist Degree Centrality  Artist Closeness Centrality
3140     Pablo Picasso                  0.010245                     0.009617
3139      Paul Cezanne                  0.007819                     0.007471
3077         Rembrandt                  0.006740                     0.002330
3151   Gustave Courbet                  0.006740                     0.003590
3076        Caravaggio                  0.006201                     0.001673
3036      Paul Gauguin                  0.005662                     0.008041
3122  Vincent van Gogh                  0.005662                     0.007976
3059            Titian                  0.005662                     0.002139
3229     Henri Matisse                  0.005392                     0.007807
3048           Raphael                  0.005392                     0.002105
```

For the movement network the nodes consisted of movements and artists while the edges consisted of influenced_on where the number of influences for a movement is represented as weight on the edges. After calculating the centralities, the most influential movement is post-impressionism:

```
Most Influential Movements:
                             Node  Movement Degree Centrality  \
67             Post-Impressionism                    0.517241
54                  Expressionism                    0.482759
74                     Surrealism                    0.408046
76                  Impressionism                    0.396552
50                       Realism                    0.373563
47                      Symbolism                    0.344828
42                    Romanticism                    0.333333
71                         Cubism                    0.321839
94                   Abstract Art                    0.316092
58  Mannerism (Late Renaissance)                     0.298851

    Movement Closeness Centrality
67                       0.131781
54                       0.166852
74                       0.173863
76                       0.123890
50                       0.138857
47                       0.120992
42                       0.113679
71                       0.124636
94                       0.181488
58                       0.081455
```

For the Institution network the nodes consisted of artists and institutions while the edges consisted of influences_on where the number of influences for an institution is represented as weight on the edges. After calculating the centralities, the most influential institution is École des Beaux-Arts:

```
Most Influential Institutions:
                                              Institution  \
4                                   École des Beaux-Arts
24  UCL Slade School of Fine Art (University Colle...
5                                         Académie Julian
20                     Art Students League of New York
28                                          Self-taught
13  Akademie der Bildenden Künste München (Munich ...
16                                Académie des Beaux-Arts
1   National Academy Museum and School (National A...
23   National Academy of Visual Arts and Architecture
15      School of the Art Institute of Chicago (SAIC)

    Institution Degree Centrality  Institution Closeness Centrality
4                        0.428571                         0.423457
24                       0.365079                         0.409797
5                        0.365079                         0.428215
20                       0.349206                         0.428215
28                       0.317460                         0.423457
13                       0.269841                         0.384961
16                       0.269841                         0.392898
1                        0.238095                         0.405437
23                       0.238095                         0.423457
15                       0.238095                         0.418803
```

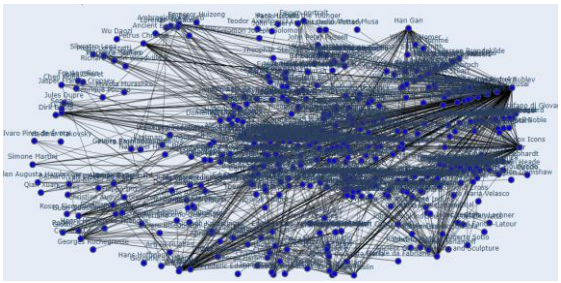
The nationality network had nodes for nationalities and artists while the edges consisted of the nationalities for the artists. To find out what nations concentrate the majority of artists I calculated the total number of edges in the network and looked at which top nations contributed to >50% of those edges. The five nations that concentrate the majority of the artists are America, France, Italy, Britain, and Germany:

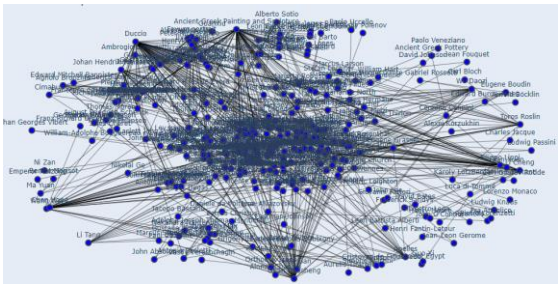
```
Total edges in the nationality network: 2996
Nations contributing to over 50% of the total connections:
American: 520 edges
French: 402 edges
Italian: 269 edges
British: 249 edges
German: 160 edges
Total edges contributed by these nations: 1600
Percentage of total edges contributed: 53.40%
```

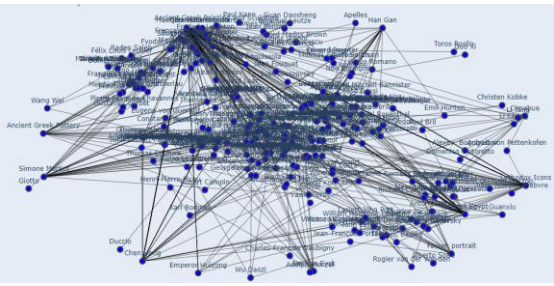
To get the top communities I used the louvian community analysis method in python which gave the node counts for the top three communities as following: 586, 378 and 342.

```
Biggest Communities in the Network:
 3071    586
3081    378
3079    342
3080    313
3077    300
3082    213
3072    213
3085    182
3075    180
3078    141
```

The top three communities are plotted using plotly below:







Conclusion

In conclusion, this project showed how network analysis can be applied across multiple tables in order to gain insight on the influence between artists, movements, and institutions in the history of art. The main bottleneck I encountered was trying to only use one network with all the nodes and edges needed to answer the different questions. I was not able to filter the output of the network sufficiently in order to get reliable answers, so I chose to create individual networks to answer all the questions which seemed to work fine.